Online Resource 1 (Electronic Supplementary Material) — A Multi-Seed Evaluation of Augmentation and Loss Interventions in Fine-Tuning Real-Time YOLO Detectors: A Study on PASCAL VOC and KITTI

Ahmed Gürkan ÖZÇELİK[1*] and Ufuk ASIL[1]

[1*]Ostim Technical University, OSTİM, 100. Yıl Bulvarı, 55/F, 06374, Yenimahalle, Ankara, Türkiye.

*Corresponding author(s). E-mail(s): 250801017@ostimteknik.edu.tr;
Contributing authors: ufuk.asil@ostimteknik.edu.tr;

Abstract

This document is Online Resource 1 (Electronic Supplementary Material) of the article titled above. It is also available in the GitHub repository. Section A gives the run log, and Section B the additional VOC results. Section C gives the additional KITTI results, and Section D the method details and computation notes. Lettered numbers such as Section B.1, Table B1, Fig. C1 and Eq. D1 belong to this document. Unlettered section, table, figure and equation numbers belong to the main article, as do references to "Appendix A of the main article". Code, logs and this supplement are available at https://github.com/ahmetgozcelik/voc-yolo-augloss-analiz. The version reported in the article is archived at https://doi.org/10.5281/zenodo.23797459.

Appendix A Run log

This section gives the full VOC run log in Table A1. Appendix A of the main article summarizes this log. Fig. A1 shows the GPU time per round and the Δ of the highest training card. The application of the stopping rule, the post-unlock work and the GPU budget are in Appendix A of the main article.

Table A1 Experiment log: number of run stamps per round and card, evidential stage-3 runs and GPU time [h]. There were 60 runs in total.

Round	Card / group	Runs	Stage 3	GPU [h]	Decision
Round 0	baseline	8 (1 invalid)	3	9.20	GREEN
	calibration	3	0	0.10	gate
Round 1	H01	3	1	3.47	CLOSE (screening, seed 0)
	H02	3	1	3.17	CLOSE (screening, seed 0)
	control (mosaic always on)	2	0	0.19	control
	H03	5	3	9.01	CLOSE (3-seed mean)
Round 2	H04	3	1	3.10	CLOSE (screening, seed 0)
	H05	4 (1 invalid)	1	3.75	CLOSE (screening, seed 0)
	H06	3	1	3.11	CLOSE (screening, seed 0)
	H07	1	1	0.19	GAIN-B (TTA)
Round 3	H08	3	1	3.16	CLOSE (screening, seed 0)
K5	baseline (test)	3	3	0.06	evaluation
	H03 (test)	3	3	0.05	evaluation
	TTA (test)	3	3	0.09	evaluation
Extra seeds	H01	2	2	6.56	screening confirmed (n=3)
	H02	2	2	5.93	screening confirmed (n=3)
	H04	2	2	5.81	screening confirmed (n=3)
	H05	2	2	6.71	screening confirmed (n=3)
	H06	2	2	5.84	screening confirmed (n=3)
	H08	2	2	5.89	screening confirmed (n=3)
	H07 WBF diagnostic	1	1	0.04	diagnostic
Total		**60**	**35**	**75.45**	target 75 h, ceiling 90 h

Runs: smoke-test, short-pilot and full-stage stamps. Invalid: an orphaned process (rc=96) and H05 stopped by the safety rule (rc=95). These count toward time but not toward evidence. K5: training-free test. Extra seeds: validation only. Because of rounding, row sums may differ from the total by ±0.02 h. Totals are computed from unrounded times.

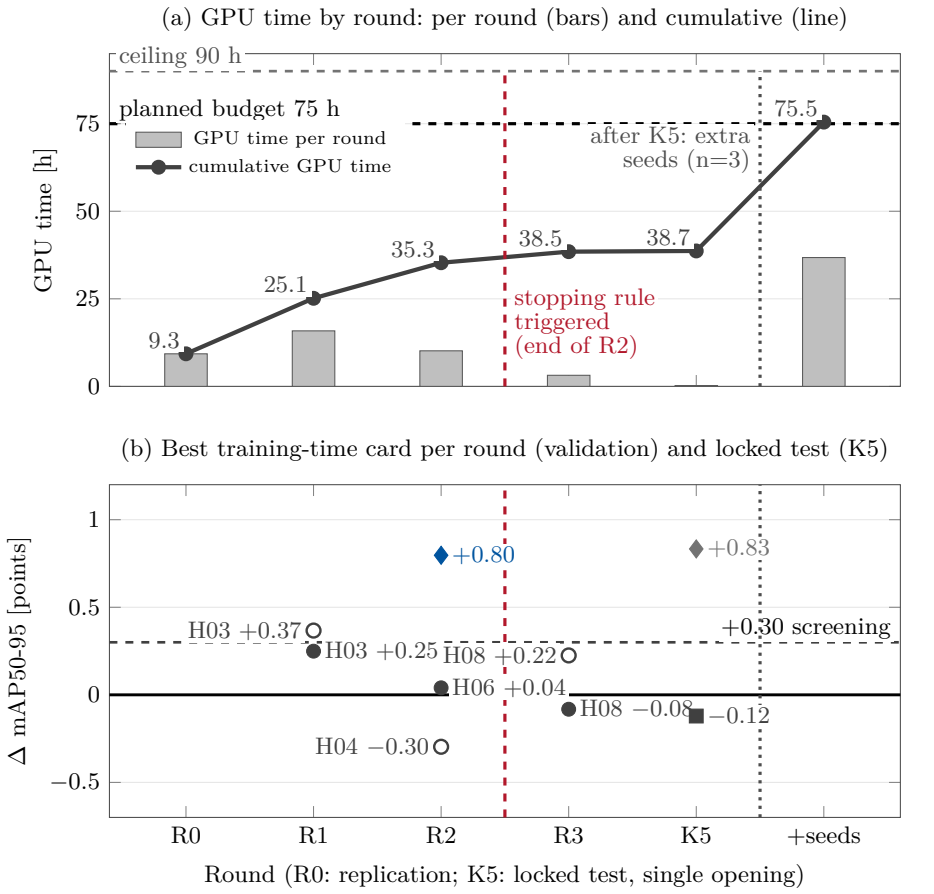


Figure A1 (a) GPU budget. (b) Δ mAP50-95 [points] vs. baseline of the best training card per round. Filled circle: 3-seed mean (with post-unlock seeds in R2–R3). Open circle: seed-0 screening. Diamond: TTA. Square: K5 test. Red line: stopping rule. "+seeds": extra seeds.

Appendix B Additional VOC results

This appendix gives the mechanism notes for the other cards. It also gives two method details: the box-based copy-paste steps, and the H07 Soft-NMS definition and confirmation procedure. Finally, it complements the VOC findings of the main text. It reports absolute values, secondary metrics, the H07 rows, the error-type breakdown, the copy-paste dose comparison and the qualitative panel. All secondary metrics here are exploratory.

B.1 Mechanism notes for the other cards

The main text reduces each of these mechanism notes to one sentence; this subsection gives them in full. All values are on VOC validation, in points, as seed-paired Δ with the three-seed mean $\pm$ s.d.

Scale axis: H01 and H04.

H01 varied the training input between 480 and 800 px per batch. All three seeds used 11 distinct sizes. H04 narrowed the mosaic scale range, and all 97 200 batches were affected in every seed. The intervention was thus active in both cards, but neither raised mAP50-95. H01's seed signs are mixed. All three of H04's seeds are negative, and its interval includes zero. In the 5-epoch short pilot (seed 0), H04 was +0.56 ahead of the baseline. At 100 epochs it turned to -0.85 (seed 0: -0.30). The short pilot is therefore a code health gate that does not predict the final ranking, although this is a single-card example. The amplitude was not increased. Hence one cannot claim that "the default scale amplitude is optimal".

Box loss: H02 and H06.

H02 used Inner-CIoU with $r = 0.75$. The IoU $-$ Inner-IoU difference that enters the loss averages 0.027 over the three seeds, so the term is active. Even so, mAP50-95 dropped in all three seeds ($p = 0.244$ after Holm-7). The targeted AP_{75} did not rise either (-0.42 ± 0.27). H06 raised the box loss gain from 7.5 to 10. It did not change AP_{75} ($+0.07 \pm 0.30$). At epoch 100, the baseline's loss shares are 30.1% for box, 23.5% for classification and 46.3% for DFL. DFL has the largest share, but its weight was not tested. Thus neither the form nor the weight of the box-IoU term closed the AP_{75} gap. Assignment, the quality target or DFL are candidate explanations. Inner-IoU has reported gains [1]. The assumption that these gains would transfer to a COCO-pretrained, TAL-assigned fine-tuning was not supported.

Copy-paste dose: H03 and H08.

H03 applied pasting to 11.1% of mosaics, and H08 to 47.8%. The measured doses are 0.142 and 0.671 objects/mosaic, respectively. H03's seed-0 difference passed the screening threshold. However, its three-seed mean stayed below the KEEP criterion. Its mAP50 gain was positive in all three seeds, while its AP_{75} difference is -0.32 ± 0.18. H08 widened the dose and the pool together. With this change, the mAP50 gain fell from $+0.74 \pm 0.16$ (H03) to $+0.25 \pm 0.14$ (H08). No increasing effect could be shown (Fig. B2). The two variables were not disentangled. The "saturated mechanism"

interpretation is therefore only a weak hypothesis. The mAP50 trace seen on validation does not appear on test (Section 5.3). In this regime, multiplying small-object instances did not close the gap that appears to be related to resolution (Section 5.2).

H05 details.

H05's validation mAP50-95 loss is -2.87 ± 0.33 points. The VFL gradient norm is 0.34 of that of BCE. The per-seed values are 0.342, 0.341 and 0.341, and the loss value is 0.031. Appendix A.5 of the main article gives the definition. The implementation passed two unit tests. First, with $\alpha = 1$, $\gamma = 0$ and $y = 0$, the loss matches BCE exactly. This setting covers negative cells only, since positives are still weighted by t; it is the definition of H05-pos. Second, the $w = t$ weight and the gradient of the positive branch were checked in double precision. The check used a hand-written reference implementation. A safety threshold stopped the first run. This threshold had been inherited from H02's BCE curve: it was the baseline's epoch-5 value minus 0.03.

B.2 Box-based copy-paste steps

The transform (Section 3.2) runs after mosaic and before RandomPerspective. It uses the intersection over area (IoA), the ratio of the intersection to the area of the first box: $\mathrm{IoA}(b_j, b_l) = |b_j \cap b_l|/|b_j|$. With probability p, the following steps are applied:

1. The candidate set $\mathcal{A} = \{\, j : \sqrt{A_j} \le \tau_A,\ \max_{l \neq j} \mathrm{IoA}(b_j, b_l) \le 0.1 \,\}$ is built. Here A_j is the pixel area of the box on the mosaic canvas. The IoA condition prevents parts of other objects from entering the patch.
2. $k \sim U\{1, 2, 3\}$ is drawn, and $\min(k, |\mathcal{A}|)$ objects are selected at random from $\mathcal{A}$.
3. Each object is placed without rescaling inside the central $s \times s$ window that remains after RandomPerspective. The location must not intersect any existing box, including newly pasted ones. At most 10 attempts are made. If no location is found, the candidate is skipped.
4. The pixel patch is pasted with 3-pixel alpha blending at its edges. The idea comes from [2]; the 3-pixel width is this study's choice. The class label is kept, and a new box with the size of the source box is added.

B.3 H07: Soft-NMS definition and confirmation procedure

The Soft-NMS rows (S) use the Gaussian rule per class [3]. When a box b_i overlaps the kept box m, its score is decayed as

$$s_i \leftarrow s_i \exp\!\big(-\mathrm{IoU}(m, b_i)^2/\sigma\big), \qquad \sigma \in \{0.3,\, 0.5\}. \tag{B1}$$

The candidate pool consists of the top 1000 detections with score ≥ 0.001, without pre-NMS.

The validation set is sorted by file name and split into two halves by even and odd index (A and B, 500/500). For each family, the candidate is the best row that gives $\Delta > 0$ relative to R0 on half A. The candidate counts as CONFIRMED if it meets all three of the following conditions: (a) $\Delta \ge +0.30$ on half B. (b) On the full validation set, an image-level paired bootstrap ($B = 200$) gives a one-sided 90 percent

lower bound above zero. (c) On the baseline's seed 1 and seed 2 weights, the same row has a mean $\Delta \geq +0.20$, and both values are positive. Section 3.2 defines the latency classification (GAIN-A/B).

B.4 Absolute values and seed 1–2 means

Table B1 VOC absolute mAP50-95 [points] (mean ± s.d., n=3) and mean Δ over seeds 1–2. Color: Holm-significant only (red: worse).

Method	Validation mAP50-95	Test mAP50-95	Δ seeds 1–2
Baseline	63.07 ± 0.41	63.49 ± 0.11	–
H01 multi_scale	62.86 ± 0.20	–	−0.35
H02 Inner-CIoU	62.72 ± 0.32	–	−0.41
H03 copy-paste	63.32 ± 0.17	63.37 ± 0.20	+0.19
H04 scale	62.22 ± 0.41	–	−1.13
H05 Varifocal	*60.20 ± 0.17*	–	−2.86
H06 box loss gain	63.11 ± 0.55	–	+0.27
H08 high dose	62.99 ± 0.14	–	−0.24
Baseline + TTA (training-free)	–	64.32 ± 0.11	–

Δ seeds 1–2: independent mean excluding the screening seed s0 (validation). The test was opened only for the baseline, H03 and TTA. Color criterion: Holm-7 on validation; on test, Holm-12 and a CI that excludes zero (Table 4).

Table B1 gives the absolute validation and test mAP50-95 values. It also gives the seed 1–2 mean, which excludes the screening seed. The baseline scores 63.07 ± 0.41 points on validation and 63.49 ± 0.11 on test. With TTA, the test value is 64.32 ± 0.11. Among the screened-out cards, no independent seed 1–2 mean exceeds +0.30. The closest is H06 at +0.27, although it gave −0.42 on seed 0. H06 was eliminated on seed 0, yet it is the card closest to the threshold on the independent seeds. This is a reminder that the single-seed gate carries a false-negative risk (Section 5.1).

B.5 Test secondary metrics

Table B2 gives the test secondary metrics of H03 and TTA. Some differences give an uncorrected $p < 0.05$. These are AP_M and AP_L for H03, and mAP75, AP_M and AP_L for TTA. However, in the twelve-test family the smallest p after Holm is 0.112. These differences are therefore read as directional only. For TTA's test mAP50-95 difference, the seed-level t interval is [+0.29, +1.38].

On validation, H03's mAP50 gain is positive in all three seeds (+0.74 ± 0.16, $p = 0.015$). On test, it is −0.02 ± 0.21. Since 70% of the validation set comes from VOC2012, a source difference was tested. The mAP50 difference is +0.73 ± 1.23 on the VOC07 subset and +0.64 ± 0.58 on VOC12 (303/697 images). This hypothesis was not supported. `best.pt` selection optimism was not tested.

Table B2 VOC secondary metrics: seed-paired Δ vs. baseline [points], n=3. p: two-sided paired t-test. Holm-12: family of 12 tests.

Comparison	Metric	Δ mean±s.d.	p	Holm-12 p
H03 box copy-paste (test)	mAP50	-0.02 ± 0.21	0.883	1.000
	mAP75	-0.11 ± 0.13	0.268	1.000
	AP_S	$+0.81 \pm 2.55$	0.639	1.000
	AP_M	$+0.48 \pm 0.14$	$0.027^{\dagger}$	0.222
	AP_L	-0.23 ± 0.09	$0.048^{\dagger}$	0.333
Baseline + TTA (test)	mAP50	$+0.10 \pm 0.27$	0.579	1.000
	mAP75	$+1.12 \pm 0.30$	$0.023^{\dagger}$	0.222
	AP_S	-0.48 ± 0.82	0.416	1.000
	AP_M	-1.07 ± 0.18	$0.009^{\dagger}$	0.112
	AP_L	$+1.50 \pm 0.36$	$0.019^{\dagger}$	0.210
H03 (validation)	mAP50	$+0.74 \pm 0.16$	0.015	–

mAP50-95 rows: Table 4. Holm-12: the K5 family (H03, TTA × 6 metrics; H03 mAP50-95 included, conservative). The smallest value is 0.112, so no color is used. †: uncorrected $p < 0.05$, not significant after Holm.

B.6 H07 training-free inference rows

Table B3 gives H07's twelve training-free rows relative to the R0 reference, together with their latency. It also gives the TTA test row. Fig. 3 plots the same rows (Section 5.2). T1 is the only CONFIRMED row. Because of its latency ratio, it counts as GAIN-B. Two additional details of the WBF diagnosis follow. First, WBF's IoU threshold is looser than that of NMS. As a result, even a single branch departs from raw NMS at a threshold of 0.6. Second, the default WBF setting also lost at high IoU. Relative to T1, AP_{75} changed by -1.33 and AP_{90} by -2.15 (Section B.7).

B.7 Training-free inference: WBF diagnosis

The default WBF setting lowered T1's mAP50-95 by -1.08. An exploratory diagnosis was therefore carried out on validation, after the test evaluation. It is not a gain claim. The implementation was correct. All 8 unit tests passed. With an IoU threshold of ≈ 1, single-branch WBF deviated from raw NMS by $+0.000$ points. The loss comes from two settings. The first is mean score fusion. With it, a box seen in fewer than three views gets a smaller score. The scale branches cannot see objects of every size, so this penalty disrupts the ranking. Max fusion brought $+0.98$ points. The second setting is the IoU threshold. Raising it to 0.7 reduced the loss under mean fusion as well. The contributions of the two settings were not separated. The best grid setting (IoU 0.7, max fusion) exceeded T1 by $+0.45$. It was selected on validation, so it is not evidence. It only indicates that the effect depends on the setting.

Table B3 H07 training-free inference rows (validation, seed 0) and the TTA test row: Δ mAP50-95 and latency. Blue: confirmed. Red: $\Delta \leq -0.5$ or $>1.10\times$.

Row	Description	Δ mAP50-95	Δ AP_S	Δ AP_L	[ms]	[×]	Status
R0	Reference (NMS 0.7, 640 px)	0.00	0.00	0.00	8.45	1.00	reference
N50	NMS IoU 0.5	*−0.54*	−0.93	−0.56	8.25	0.98	not a candidate
N60	NMS IoU 0.6	−0.21	−0.20	−0.20	8.30	0.98	not a candidate
N80	NMS IoU 0.8	−0.10	+0.03	−0.03	8.33	0.99	not a candidate
S03	Soft-NMS σ=0.3	−0.12	−0.11	−0.03	9.10	1.08	not a candidate
S05	Soft-NMS σ=0.5	*−0.58*	+0.06	−0.51	9.14	1.08	not a candidate
U704	test imgsz 704	−0.31	+2.41	−0.82	8.92	1.06	not a candidate
U768	test imgsz 768	*−2.06*	+4.55	−3.41	9.43	1.12†	not a candidate
U832	test imgsz 832	*−5.19*	+4.94	−7.24	9.61	1.14†	not a candidate
T1	TTA 1, 0.83, 0.67 + flip	**+0.80**	−0.12	+1.58	21.68	*2.57*	GAIN-B
T2	TTA 1, 1.2, 1.4 + flip	*−4.12*	+2.75	−5.99	21.54	*2.55*	not a candidate
W1	WBF (T1 branches)	−0.28	+0.41	+0.65	25.27	*2.99*	not a candidate
W2	WBF (T2 branches)	*−5.74*	+5.10	−7.85	26.14	*3.10*	not a candidate
T1 test	TTA, 3 seeds, test	+0.83 ± 0.22†	−0.48	+1.50	21.64	*2.29*	GAIN-B

Validation: baseline seed-0 `best.pt` (R0 = 8.45 ms). TTA test: its own session (baseline 9.43 ms). Red Δ: negative in both validation halves (no CI). †: ratio within single-run noise (1.14×), left uncolored; or Δ with uncorrected $p < 0.05$.

B.8 Per-card secondary metrics (validation)

The training cards show the following differences in the size and threshold metrics. For H01, AP_S is -0.30 ± 2.24 and AP_M is $+0.84$. For H04, AP_S is -1.06 ± 0.68 and AP_M is -0.75. For H02, the AP_S change is $+1.39 \pm 1.09$. For H06, mAP50 is $+0.30 \pm 0.55$. All these AP_S changes lie within the baseline's seed range of 2.87 points. H05's loss appears in all three size classes: AP_S −2.39, AP_M −4.06 and AP_L −2.10.

Over the last 10 epochs, H05's validation gain is +0.20, +0.27 and +0.23 points. The baseline's gain is +0.04, +0.16 and −0.03. The difference is ≈0.18 points. This is far below the noise of a single validation measurement (Section 5.3). This is also the phase in which mosaic is turned off. The convergence hypothesis therefore receives only weak support.

The per-seed values of H05c are as follows. $\Delta_{c,s}$ is +0.02, −1.15 and −0.40. D_s is +2.90, +2.04 and +2.13. The per-seed recovery is 1.01, 0.64 and 0.84, with a mean of 0.83. In the remaining difference, two seeds are negative and one is ≈0. The VFL/BCE gradient norm ratio measured in the H05c runs is 0.347–0.349. The best epoch is 96, 96 and 95 for H05c, and 100, 100 and 100 for H05. The decision rule fixed two cases in advance. An invalid run would not be rerun; it would be reported as "could not be

run". If magnitude did not explain the loss, this would strengthen the form hypothesis but not prove it.

B.9 Error-type breakdown

Detections are divided into error types in a manner similar to TIDE [4]. Matching uses pycocotools with at most 300 detections. The types are defined as follows:

- A true positive (TP) is a detection matched with IoU≥ 0.5.
- Dupe: a detection with IoU≥ 0.5 whose GT has already been taken by a higher-scoring box.
- Cls: IoU≥ 0.5 with a GT of another class.
- Loc: $0.1 \leq$IoU< 0.5 with a GT of the same class.
- Both: $0.1 \leq$IoU< 0.5 with a GT of another class.
- Bkg: a detection with IoU≥ 0.1 to no GT.
- Miss: a GT that no box matches with IoU≥ 0.5.

Oracle ΔAP_{50} is the increase in AP_{50} when the given error is fixed and the detections are re-evaluated. The fixes are: Loc, the box is snapped to the GT; Cls, the class is corrected; Dupe, Both and Bkg, the box is deleted; Miss, the GT is deleted. The breakdown is reported at the operating point (score ≥ 0.25) and without a threshold (≥ 0.001). It is also reported at the size and class level.

Fig. B1 compares the oracle ΔAP50 cost of each error type for the baseline, H03 and H06. The test breakdown is exploratory. H06 is shown on val only, for seed 0. Background false positives (Bkg) are the costliest type on both sets, at 3.48 points on val and 3.79 on test. Miss costs 3.32 and 2.03, respectively. The H03 and H06 bars deviate from the baseline by at most 0.31 points on val and at most 0.06 on test. The error profile thus did not shift. This is consistent with the near-zero difference in mAP.

B.10 Copy-paste dose comparison

Fig. B2 plots the validation gain against the paste dose in objects per mosaic. The doses are 0 for the baseline, 0.142 for H03 and 0.671 for H08, which is 4.7 times that of H03. H08 changes the probability and the candidate pool together (Section 3.2). Dose and size mix are therefore not disentangled. The figure is a three-point comparison, not a dose–response curve. ΔmAP50 falls from $+0.74 \pm 0.16$ to $+0.25 \pm 0.14$. For H08, ΔAP_S is $+0.31 \pm 2.84$. When dose and pool were widened together, the gain did not grow ("no increasing effect could be shown").

B.11 Qualitative panel

Fig. B3 shows GT, baseline, H03 and baseline+TTA detections on the same test images (seed 0, score ≥ 0.25). It uses extreme examples in both directions and one random frame. The caption gives the selection criterion. The pool has 1642 images, and the baseline image-AP Q1–Q3 range is 0.497–0.896. In one scene, TTA raises image-AP from 0.522 to 0.971. In a car scene, it lowers image-AP from 0.895 to 0.248. H03 also jumps in the first scene, so the differences are not specific to TTA. Over all valid images (seed 0), TTA's mean image-AP difference is -1.26. Of the images,

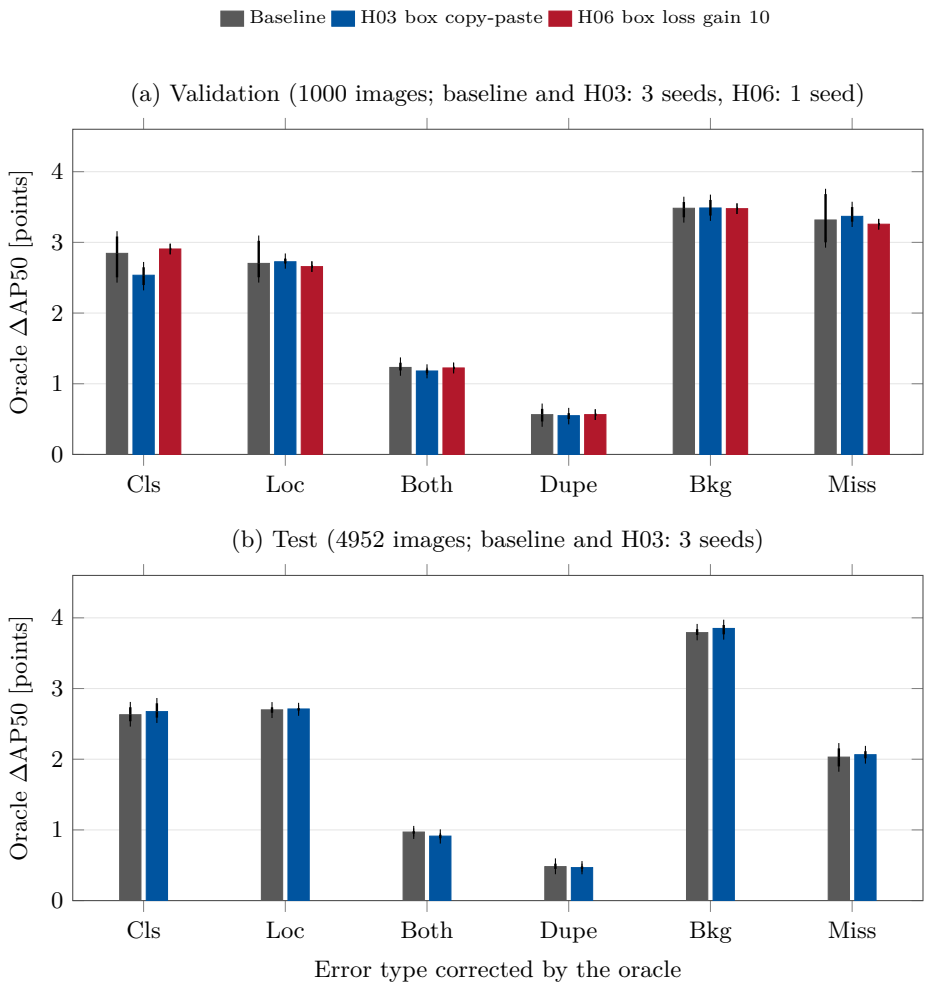


Figure B1 Error-type oracle: ΔAP50 [points] if the given error is corrected (Cls classification, Loc localization, Both both, Dupe duplicate, Bkg background, Miss missed). Bars show the mean and whiskers the seed min–max. (a) Validation, (b) test.

31.8% improve and 38.4% worsen. For H03, the mean difference is +0.36, with 35.5% improving and 34.7% worsening. Dataset mAP, however, increases by +0.88. The gain is thus not a uniform visual improvement (exploratory).

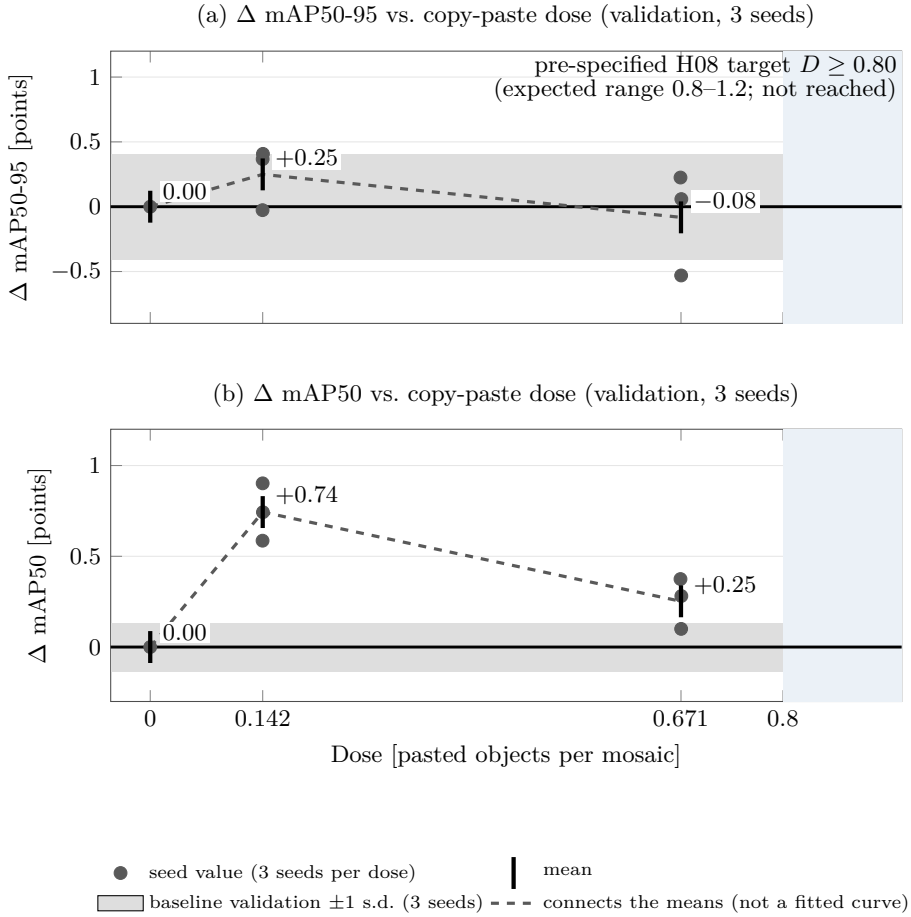


Figure B2 Validation Δ mAP50-95 (a) and Δ mAP50 (b) vs. box-based copy-paste dose (pasted objects per mosaic), 3 seeds. H08 changes the paste probability and the candidate pool together. The effect of dose alone is not isolated.



Figure B3 GT, baseline, H03 and TTA detections on six VOC07 test images (seed 0). Bold blue/italic red: better/worse than the baseline. Candidates had ≥2 GT objects, detections from all three methods and a baseline AP within Q1–Q3. From these we selected a TTA gain and a TTA loss, an H03 gain and an H03 loss with small objects, and 1 random image.

Appendix C KITTI supplementary results

This section gives the setup details of the KITTI experiment (Section 7 of the main article) and its secondary results. It also covers the training curve of H05, the analysis of `best.pt` selection, the post-unlock robustness analysis, and the details of training-free inference.

C.1 Setup details

Questions, rules, and training details.

S1 is answered per model with two families. The improvement family counts the cards that meet the acceptance criterion. The directional replication family asks whether the loss seen on VOC is replicated. S2 is answered with the contribution and additivity criteria of the two component cards run on VOC. Later changes to the rule document do not apply to the VOC cards of S2. The actual optimizer behavior on VOC was fixed with the following settings: MuSGD, learning rate 0.01, momentum 0.937, bias warmup 0. The copy-paste parameters were unchanged. The number of objects pasted per mosaic is 0.98 in H03 and 2.00 in H08. In YOLO26n, the H02 and H05 patches are applied to both training heads; a head counter verified this. With three seeds, the critical effect at 50% power in the first step of Holm-5 is $\approx 2.8\, s_{\Delta}$. For a card that is not accepted, we state "acceptance criterion not met". In H07, the NMS, Soft-NMS, and WBF rows are not applicable to YOLO26n, which is NMS-free.

Split and ignore rule.

We did not use the ready-made Ultralytics split. Nor did we use the 3712/3769 train/validation split common in 3D detection work. That split has no separate test partition. We also could not verify that it includes a check (NCC) that groups frames of the same scene stored under different recording names. Scene groups were assigned to the training, validation, and test partitions under two constraints. At least 40%, 5%, and 10% of each class's objects fall in the respective partition. No single group exceeds 25% of the target validation or test size. The highest cross-partition NCC is 0.855 for test–train and 0.888 for validation–train. The test partition contains 6 212 objects and 2 436 `DontCare` regions. In evaluation, `DontCare` regions are added for every class as COCO-style ignore regions (`iscrowd`). Detections are first matched to real objects. Only an unmatched detection can be ignored. It is ignored if its overlap with a `DontCare` region is greater than or equal to the current IoU threshold. Overlap is measured as IoA, the intersection divided by the detection area. The official KITTI AP, by contrast, uses a fixed per-class threshold and difficulty levels. mAP without `DontCare` is reported as a secondary result in Table C1.

Training, health gate, and learning check.

The loss code of YOLO26n is v8.4.165 `E2ELoss`. The number of iterations on KITTI is ≈3.1 times lower than on VOC (24 300). The head counter was read in every H02 and H05 run. A training run took 1.68–1.83 GPU hours for YOLO11s and 1.17–1.31 for YOLO26n. Every run passes an automatic gate with three parts. The batch must not be silently reduced, the patch must actually be called, and a learning check must pass.

The learning check is a validity criterion with five conditions: `results.csv` contains exactly 100 epochs; loss and metric values are finite; the learning rate schedule matches that of the seed-0 baseline of the same model, epoch by epoch; the mean training loss (box + classification) over epochs 81–90 is lower than over epochs 11–20; and the best validation mAP50-95 is above 0.02. No threshold relative to the baseline was set, because it would censor strongly negative cards. A run that fails does not enter Δ. A best epoch below 20 is a warning, not an invalidation. The rule is applied identically to all cards and both models. Two rule changes were made: this check was added to the rule document, and `last.pt` evaluation was added to the test script (Section 9 of the main article). Until the YOLO11s queue finished, monitoring used only the event type, run ID, and exit code.

Resolution.

Most KITTI frames are 1242×375 (1 396/1 504 in the test set), with an aspect ratio of 3.31. At 640 px input, the long side is scaled by ≈0.52, so a frame becomes ≈640×193. In the test set, 19.1% of pedestrian and cyclist objects fall in the COCO "small" category (area $< 32^2$ px) at native resolution. At 640 px, this share is 65.6%. VOC images, in contrast, are at most 500 px and are upscaled by a factor of 1.28 at 640 px. As a result, the small-object share in the VOC test set drops from 4.7% to 1.85%. Following the single-variable rule, the training resolution was kept the same as for VOC (640 px). This difference is the simplest hypothesis for two contrasts. First, 832 px test input yields a gain on KITTI but a loss on VOC. Second, multi-scale training (H01) is accepted on KITTI but not on VOC. The reasoning is that KITTI has a high small-object share, so a higher effective resolution may help. On VOC, the objects are already upscaled at 640 px. The hypothesis was not tested.

Baseline plausibility.

No published anchor exists for KITTI with this split and evaluation rule. The VOC plausibility gate was therefore not applied, and three checks were made instead. The first is a training pipeline diagnosis. The baseline `last.pt` weights give a high mAP50-95 on training images and a low one on validation (numbers in Section C.4). The second check is the absolute values, in points, with the `DontCare` rule and averaged over three seeds. Test mAP50-95 is 19.30 (`best.pt`) and 19.66 (`last.pt`) for YOLO11s, and 18.85 and 17.35 for YOLO26n. The `best.pt` validation mAP50-95 is 19.26 ± 2.10 for YOLO11s and 18.51 ± 1.49 for YOLO26n. Test car AP50-95 (`best.pt`) is 57.66 for YOLO11s and 58.10 for YOLO26n. These values only indicate the scale of the Δs. They are not a comparison anchor, and they are not compared with the VOC values or across models. The third check is an external reference value, used for order of magnitude only (Section D.6). Its resolution, split, and evaluation differ from ours, so it is not comparable with our values.

C.2 KITTI secondary results and best epoch per run

This subsection gives the secondary reports defined in the KITTI rule document. All are descriptive. There is no Holm correction and no significance claim.

Table C1 KITTI pre-specified secondary descriptive comparisons: card − baseline Δ, mean ± s.d. [points], $n = 3$ seed-paired. No Holm correction, no color.

	Test Δ			Val Δ
Card	No DontCare	3 classes	7 classes	ep11–90 curve
YOLO26n				
H01 multi-scale	+0.05 ± 2.19	+1.22 ± 1.18	+0.08 ± 2.52	−0.31 ± 0.03
H02 Inner-CIoU	−1.01 ± 1.98	+1.04 ± 1.17	−1.02 ± 2.29	+0.51 ± 0.52
H03 box copy-paste	+0.88 ± 0.68	+2.81 ± 0.65	+1.07 ± 0.87	+0.62 ± 1.02
H04 mosaic scale	−1.33 ± 2.19	−0.97 ± 1.38	−1.53 ± 2.46	−0.54 ± 0.48
H05 Varifocal	+0.47 ± 1.48	+1.28 ± 1.16	+0.78 ± 1.73	+1.29 ± 0.20
H06 box loss gain	−0.24 ± 0.72	+0.56 ± 1.68	−0.26 ± 0.86	+0.33 ± 0.71
H08 high dose	+0.67 ± 0.68	+2.68 ± 1.32	+0.73 ± 0.89	+1.01 ± 0.83
YOLO11s				
H01 multi-scale	+1.48 ± 0.39	+0.48 ± 1.72	+1.72 ± 0.29	+0.59 ± 0.57
H02 Inner-CIoU	+1.13 ± 0.48	+0.06 ± 1.75	+1.38 ± 0.66	−0.27 ± 0.55
H03 box copy-paste	+1.17 ± 1.00	+0.22 ± 2.73	+1.50 ± 1.17	+1.01 ± 0.55
H04 mosaic scale	+1.00 ± 0.83	−0.66 ± 1.86	+0.86 ± 1.30	−0.36 ± 1.36
H05 Varifocal	+2.35 ± 0.96	+1.41 ± 2.06	+2.73 ± 1.60	+0.53 ± 0.37
H06 box loss gain	+0.45 ± 0.65	+0.13 ± 1.31	+0.57 ± 0.79	−0.38 ± 0.29
H08 high dose	+1.63 ± 2.89	+1.30 ± 4.20	+1.66 ± 4.04	+1.00 ± 0.39

Test (`best.pt`): mAP50-95 without DontCare. Subsets with the DontCare rule: car, pedestrian, cyclist (3 classes) and 7 classes excluding person_sitting. Val: mean of the training curve (Ultralytics, without DontCare) over epochs 11–90. Test object counts: person_sitting 56, tram 126, truck 119.

Table C1 shows the test Δ without `DontCare`. In none of the 14 cards does it differ from the primary Δ (with the `DontCare` rule) by more than 0.24 points. The primary result is therefore not sensitive to the ignore rule. On the standard three classes (car, pedestrian, cyclist), the gain of YOLO11s H01 is $+0.48 \pm 1.72$. Its per-seed Δs are +1.57, −1.50, and +1.38. On the seven-class subset that excludes person_sitting, the gain is $+1.72 \pm 0.29$, positive in all three seeds. The difference appears when van, truck, tram, and misc are added to the standard three classes. These classes have far fewer test objects than car (4 457): van has 464, misc 184, tram 126, and truck 119. person_sitting has 56. With these counts, per-class AP is noisy. The distribution of the gain over individual classes cannot be resolved with these data. YOLO26n gives an example in the opposite direction. Its three-class difference is $+2.81 \pm 0.65$ for H03 (positive in all three seeds) and $+2.68 \pm 1.32$ for H08. The primary Δs of these two cards are far below this (Table 5 of the main article). The validation curve column uses the mean validation mAP50-95 over epochs 11–90 during training. It reports the difference of this mean from the baseline. For H05, it is $+1.29 \pm 0.20$ for YOLO26n and $+0.53 \pm 0.37$ for YOLO11s (Section C.3).

Table C2 gives the best epoch of each run. It also gives the in-training validation difference between the best and the last epoch. For YOLO11s, the best epoch is 1 in 11 of 24 runs: 2 in seed 0, 3 in seed 1, and 6 in seed 2. For YOLO26n, this happens in

Table C2 KITTI: best epoch per run (`best.pt`) and val best−last difference [points]. 48 runs, seeds s0–s2. Italic: epoch 1.

	s0		s1		s2	
Arm	Epoch	Diff.	Epoch	Diff.	Epoch	Diff.
YOLO26n						
B baseline	16	+4.3	11	+5.1	28	+3.7
H01 multi-scale	24	+3.8	9	+4.6	20	+5.3
H02 Inner-CIoU	29	+3.8	11	+4.0	38	+3.1
H03 box copy-paste	10	+4.7	24	+4.4	37	+2.6
H04 mosaic scale	16	+5.5	11	+4.0	12	+5.4
H05 Varifocal	46	+2.9	39	+4.2	33	+2.8
H06 box loss gain	51	+2.9	11	+4.8	28	+3.4
H08 high dose	39	+3.4	12	+4.6	48	+3.3
YOLO11s						
B baseline	9	+4.4	32	+3.1	*1*	+5.7
H01 multi-scale	15	+4.6	48	+3.9	*1*	+5.5
H02 Inner-CIoU	29	+4.6	38	+4.6	*1*	+6.6
H03 box copy-paste	*1*	+4.8	12	+2.7	*1*	+5.2
H04 mosaic scale	39	+5.1	*1*	+7.4	*1*	+6.1
H05 Varifocal	47	+2.7	54	+1.9	17	+3.2
H06 box loss gain	13	+4.9	*1*	+3.6	*1*	+6.5
H08 high dose	*1*	+2.9	*1*	+6.3	24	+3.9

Val difference: in-training best − last-epoch mAP50-95 (Ultralytics, without DontCare). Runs whose best epoch is 1: YOLO26n 0/24, YOLO11s 11/24.

no run. The difference is positive in all 48 runs. This is expected, because `best.pt` is selected on validation. The difference is measured with Ultralytics' in-training evaluator, which does not use `DontCare`. The 24-run mean is +4.03 for YOLO26n and +4.58 for YOLO11s. For the baseline alone, it is +4.37 and +4.39 points. These numbers come from a different measurement than the +3.23 and +3.84 in Section C.4, which come from this study's COCO-style re-evaluation.

Table C3 gives three quantities per card. These are the secondary test metrics, the validation Δ, and the difference between the in-training evaluation and the re-evaluation. Row B gives absolute values. For YOLO11s, seven of seven cards are positive on all four metrics. The ranges are +0.80 to +2.14 for mAP50, +1.28 to +5.69 for AP_S, +0.58 to +1.60 for AP_M, and +1.28 to +4.31 for AP_L. The validation Δ is also positive for all seven cards. This fits a common difference that shifts all cards against the same baseline better than a card-specific effect (Section C.4). For YOLO26n, the signs are mixed. The number of positive cards is 4/7 for mAP50, 3/7 for AP_S, 4/7 for AP_M, and 4/7 for AP_L. The last column compares two evaluations of the same `best.pt` weights. It is the best in-training validation value minus the COCO-style re-evaluation without `DontCare`. The 24-run mean is +0.37 for YOLO26n and +0.40 for YOLO11s. Over all 48 runs, the range is −1.03 to +1.80, and 37/48 runs are positive. `best.pt` is selected by Ultralytics fitness. It therefore relies on a different

Table C3 KITTI secondary metrics (`best.pt`): card − baseline Δ, mean ± s.d. [points], $n = 3$ seed-paired. Test uses the DontCare rule. Descriptive, no color.

	Test Δ				Val [points]	
Card	mAP50	AP_S	AP_M	AP_L	Δ mAP50-95	In-train−re-eval.
YOLO26n						
B	33.59±0.89	7.92±2.61	15.94±1.05	32.71±0.94	–	+0.35±1.20
H01	+0.23±1.82	+0.05±1.21	−0.37±1.48	+2.32±3.38	−0.10±2.25	+0.21±1.01
H02	−1.31±2.54	+0.29±2.68	−0.51±2.18	−1.86±0.40	−0.29±0.80	+0.51±0.52
H03	+1.94±0.43	−0.10±1.47	+1.05±0.18	+1.81±3.11	+0.63±1.06	+0.19±0.46
H04	−1.90±2.14	−1.80±1.65	−0.83±2.08	−1.38±0.97	−0.22±1.89	+0.78±0.59
H05	+0.40±2.14	+0.13±1.82	+1.02±1.75	+0.57±2.08	+0.68±1.14	+0.44±0.53
H06	−0.38±1.10	−0.30±1.92	+0.18±1.31	−0.90±1.91	+0.09±0.65	+0.20±0.32
H08	+1.18±0.69	−1.24±3.08	+0.72±1.54	+0.96±1.53	+0.46±1.34	+0.29±0.52
YOLO11s						
B	35.39±0.65	7.20±0.91	18.14±0.80	31.92±0.86	–	+1.12±0.68
H01	+1.63±1.08	+2.58±1.61	+0.98±1.77	+2.74±2.76	+1.73±1.78	−0.01±0.33
H02	+1.87±2.46	+1.28±0.84	+0.95±0.84	+2.10±1.24	+1.15±1.05	+0.28±0.29
H03	+1.84±2.56	+2.50±2.35	+0.60±1.22	+1.28±1.94	+1.86±2.44	+0.65±0.54
H04	+2.14±0.26	+2.81±3.12	+0.81±1.34	+2.50±1.03	+1.82±0.43	+0.74±1.00
H05	+1.66±0.76	+4.37±2.90	+1.60±1.01	+2.30±1.57	+0.33±2.09	+0.47±0.33
H06	+0.80±1.62	+5.69±1.31	+0.62±0.40	+2.10±1.00	+1.24±0.56	+0.03±0.38
H08	+2.04±1.65	+1.99±1.28	+0.58±0.85	+4.31±1.98	+2.21±3.24	−0.06±0.61

Cards as in Table 5. Val Δ: mAP50-95 with the DontCare rule. In-train−re-eval.: in-training best val (Ultralytics) − COCO-style re-evaluation of the same weights without DontCare. Row B (baseline) gives absolute values. $AP_S/AP_M/AP_L$ are the COCO area classes.

criterion from the evaluator of the primary endpoint. AP_S is the COCO small-object category. For YOLO26n, the s.d.s in this column range from 1.21 to 3.08. Individual card differences should be read with this noise in mind.

Table C4 gives the per-class AP50-95 Δ. The other classes have far fewer test objects than car; the counts are given above and in the table footnote. For classes with few samples, the s.d.s can therefore exceed the differences themselves. For example, truck in YOLO11s H01 gives $+0.1 \pm 3.0$. The distribution of the gain over classes cannot be resolved with these data.

C.3 Training curve of H05 on KITTI

On KITTI, the same intervention did not produce the loss seen on VOC (Section 7.2 of the main article). Measured: H05 peaks later than the baseline in both models. For YOLO11s, the best epochs are 47/54/17 for H05 and 9/32/1 for the baseline. For YOLO26n, they are 46/39/33 for H05 and 16/11/28 for the baseline. The peak is later in 6/6 seeds. This instance is weak for YOLO11s seed 2, however, because the baseline's best epoch there is 1. The late peak is most consistent for H05. For each of the other six cards, a later peak than the baseline appears in at most 4 of the six

Table C4 KITTI test: per-class AP50-95 (DontCare rule, `best.pt`), card − baseline Δ, mean ± s.d. [points], $n = 3$ seed-paired. Descriptive, no color.

Card	car	van	truck	ped.	sit.	cyc.	tram	misc
YOLO26n								
B	58.1	12.4	13.7	24.5	1.1	7.2	33.3	0.6
H01	−0.6±2.0	+0.2±1.9	−1.6±4.0	+1.8±2.0	+0.5±0.5	+2.5±2.6	−1.5±11.0	−0.1±0.4
H02	+0.3±1.2	−1.6±1.9	−3.3±3.0	+0.4±2.4	−0.3±0.8	+2.4±0.8	−5.5±12.3	+0.1±0.3
H03	+0.5±1.3	+1.8±0.4	+1.4±2.9	+1.8±0.6	+0.6±0.9	+6.1±1.8	−4.3±5.2	+0.3±0.3
H04	−1.3±1.3	−0.5±2.5	−0.3±1.7	−2.4±2.0	+0.8±1.1	+0.7±1.3	−7.2±10.7	+0.2±0.5
H05	+1.3±0.4	+0.8±1.8	−3.2±4.3	+0.9±0.5	−0.5±0.1	+1.6±2.8	+3.1±5.9	+0.8±0.7
H06	−0.3±1.1	−0.6±1.3	−0.4±5.6	−0.1±1.2	−0.2±0.8	+2.1±3.6	−2.6±8.6	+0.1±0.3
H08	+0.7±1.6	−0.8±2.0	+1.0±4.4	+2.2±1.4	+0.6±0.7	+5.2±1.1	−3.0±5.9	−0.1±0.4
YOLO11s								
B	57.7	12.8	13.5	27.0	3.4	15.7	22.9	1.4
H01	+2.2±0.9	+2.4±0.9	+0.1±3.0	+1.8±2.4	−0.1±0.7	−2.6±3.4	+8.2±3.0	−0.2±0.5
H02	+1.0±1.0	−0.4±3.0	+1.6±3.2	+1.1±0.7	−0.6±0.7	−1.9±4.9	+7.8±5.6	+0.5±0.6
H03	+1.7±1.6	+2.8±0.5	+3.9±3.4	+1.9±4.2	+0.7±5.1	−3.0±3.2	+2.9±4.1	+0.3±1.6
H04	−0.6±2.6	+3.1±5.2	+1.1±4.5	−0.3±1.0	+2.5±4.8	−1.1±4.6	+3.5±11.8	+0.3±1.1
H05	+4.3±1.8	+2.2±2.5	+4.1±2.5	+1.7±0.5	−0.8±5.4	−1.8±4.3	+8.3±6.8	+0.2±1.1
H06	+0.0±2.1	+1.5±3.9	+3.3±5.1	+2.6±2.9	+0.6±1.2	−2.1±4.9	−1.1±10.9	−0.1±1.0
H08	+2.1±4.0	+2.6±1.9	+1.4±4.3	+4.0±2.4	+1.9±8.5	−2.2±6.3	+2.7±21.6	+0.9±1.3

Cards as in Table 5. Abbreviation: sit. = person_sitting. Row B gives absolute AP. Test object counts: car 4457, van 464, truck 119, ped. 574, sit. 56, cyc. 232, tram 126, misc 184.

seeds of the two models (H02, H08). For YOLO26n, H05 has the largest difference among the cards in the epoch 11–90 mean of the validation curve. This difference is +1.29 ± 0.20 and is positive in all three seeds. For YOLO11s, it is not specific to H05. H05 gives +0.53 ± 0.37, while H03 gives +1.01 and H08 gives +1.00 (Table C1). The mechanism was active in every run. For the remaining hypothesis and its limitation, see Section 7.2 of the main article (answer to S2).

C.4 Checkpoint selection: `best.pt` analysis

This subsection details the sensitivity to `best.pt` selection that is summarized in Section 7.2 of the main article. The analysis is exploratory. The symptom is that the YOLO11s `best.pt` Δ is positive in seven of seven cards (Fig. C1b). This includes H02 and H05, which are downward on VOC. Four candidate explanations were tested, with the outcome given below: an early peak; noise in the validation set of only 10 scene groups; the outlying baseline seed 2; and the selection of `best.pt` by validation fitness without `DontCare`, while the primary endpoint uses the `DontCare` rule.

`best.pt` and `last.pt` per card.

Fig. C1 plots the per-card `best.pt` and `last.pt` Δs side by side. For YOLO11s (b), the `best.pt` Δ is positive in seven of seven cards, including H02 and H05, which are

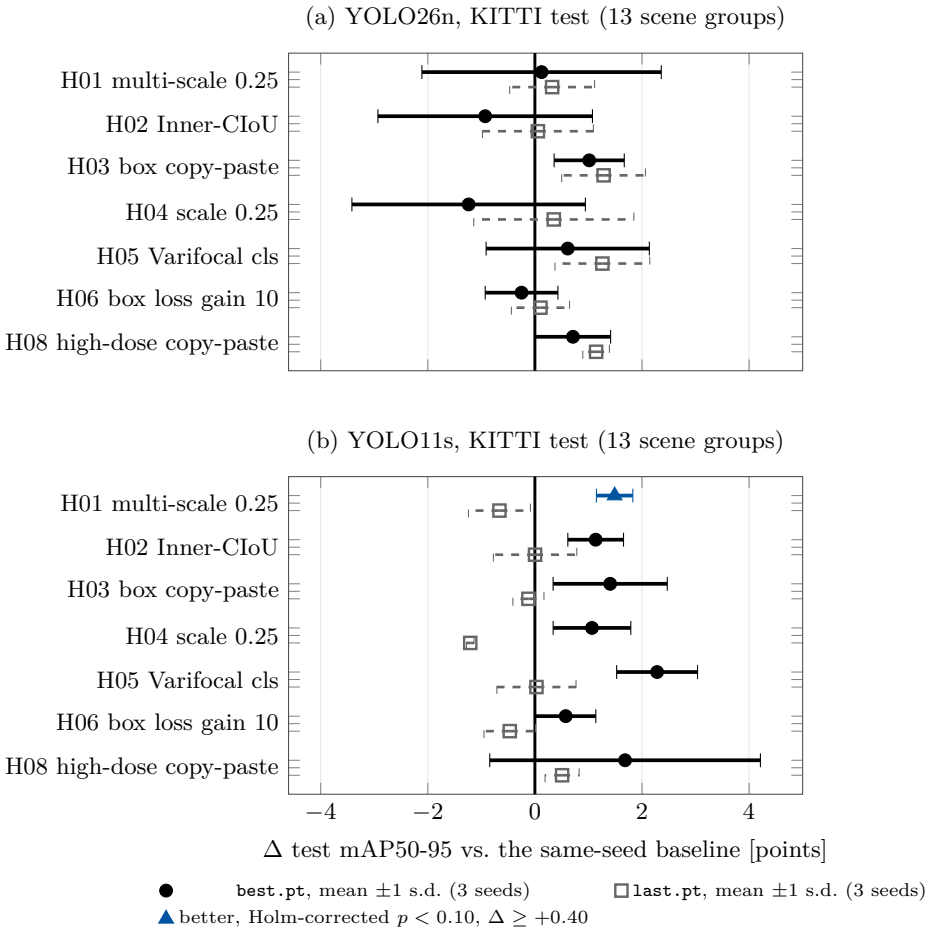


Figure C1 KITTI test: per-card Δ mAP50-95 vs. baseline [points]. Points show the three-seed mean ±1 s.d. for `best.pt` and `last.pt`. (a) YOLO26n, (b) YOLO11s. Blue triangle: Holm-5 acceptance.

downward on VOC. The `last.pt` Δs, by contrast, scatter around zero. For YOLO26n (a), the pattern is mirrored. Three cards are negative at `best.pt`, whereas all seven are above the baseline at `last.pt`. The robustness label was "sensitive to `best.pt` selection" for 7 of the 14 cards (Table 5 of the main article). For YOLO11s, H02 and H05 are labeled "selection-independent". This is a borderline case of the sign rule, because their `last.pt` Δs are ≈0. The common shift of the cards is the mean test mAP50-95 of the seven cards minus the baseline mean. For YOLO11s, it is +1.38 points at `best.pt` and −0.27 at `last.pt`. For YOLO26n, it is +0.01 and +0.65. The shift thus depends on the baseline checkpoint. The seven cards are measured against a single baseline, so 7/7 is not seven independent pieces of evidence. For YOLO11s, the clustered CI excludes zero for five cards. This too fits the common shift better than a card effect. Remaining hypothesis: the validation set is small (10 scene groups). On it, `best.pt` selection captures a peak specific to this set, and for YOLO11s it often picks

the epoch-1 weights. This peak partly does not carry over to the test set (winner's curse and scene-group noise). The other explanations and the training pipeline diagnosis are given in the paragraphs below. The experiment that would discriminate among the explanations was not run. It would require fixed-epoch evaluation at intermediate epochs, or more seeds.

Epoch-1 weights.

For YOLO11s, the best epoch is 1 in 11 of 24 runs and below 20 in 16 runs. The median is 10.5 (Table C2). For YOLO26n, no run has a best epoch of 1 (0/24), 10 runs are below 20, and the median is 24. In YOLO11s seed 2, `best.pt` is the epoch-1 weights in six of the eight arms: baseline, H01, H02, H03, H04, and H06. Five of the card–baseline comparisons are therefore between two epoch-1 weights. Their mean test Δ is +1.235, with a range of +0.634 to +1.593. These five comparisons are not independent, because they share the same baseline weights. The test mAP50-95 of baseline seed 2 is 18.22 at `best.pt` and 19.83 at `last.pt`.

Training pipeline.

We measured mAP50-95 on 1000 training images. For YOLO26n baseline seed 0, `best.pt` gives 56.06 and `last.pt` gives 78.42. The validation value of the same `best.pt` is 17.89. For YOLO11s baseline seeds 0–2, `last.pt` gives 85.13, 85.00, and 85.25. The validation value of the same weights is 14.8–15.9. The epoch-1 `best.pt` has not yet learned the training scenes: it gives 36.74 for YOLO11s seed 2. Seeds 0 and 1, whose best epochs are 9 and 32, give 49.84 and 69.93. The low validation and test levels are consistent with a generalization gap rather than a pipeline error. One possible cause is overfitting to the 36 training scene groups. Another is that the optimizer and learning rate (MuSGD, 0.01) do not suit these data; they were fixed to preserve the VOC behavior. A third possibility is that full fine-tuning degrades the pretrained features on the scene-disjoint validation and test sets. This effect has been shown under large distribution shift [5]. However, the magnitude of the shift between the KITTI splits was not measured. None of the three possibilities was tested.

Selection criterion mismatch.

The mismatch hypothesis assumes that detections in `DontCare` regions penalize late epochs. It predicts that the best−last difference narrows under the `DontCare` rule. We measured the same weights on validation with this study's COCO-style evaluator, both without `DontCare` and with the `DontCare` rule. The difference widens in 48 of 48 runs. The mean goes from +3.23 to +3.62 for YOLO26n and from +3.84 to +4.31 for YOLO11s. The hypothesis is not supported. On validation, `best.pt` beats `last.pt` in 48/48 runs. This is structural, because `best.pt` is selected on validation, and it does not discriminate between explanations. Ultralytics' in-training evaluator (without `DontCare`) is a separate measurement. `best.pt` is selected by Ultralytics fitness, not by the evaluator of the primary endpoint (numbers in Table C2). On the test set, the baseline `best.pt` − `last.pt` difference is +1.50 for YOLO26n and −0.36 for YOLO11s. On validation with the `DontCare` rule, the same difference is +3.78 and +3.43 points.

For YOLO11s, the validation peak has no counterpart on the test set. For YOLO26n, it partly carries over.

Common shift.

The common shift of the cards relative to the baseline is the mean test mAP50-95 of the seven cards minus the baseline mean. For YOLO11s, it is +1.38 points at `best.pt` and −0.27 at `last.pt`, a difference of 1.65. For YOLO26n, the pattern is mirrored: the shift is +0.01 at `best.pt` and +0.65 at `last.pt` (Fig. C1a). The common shift depends on which checkpoint represents the baseline. The seven cards are measured against a single baseline, so 7/7 is not seven independent pieces of evidence. The outlier in seed 2 does not explain the pattern on its own. For YOLO11s, seven of seven cards are also positive in seeds 0 and 1. The per-seed common shift is +1.36 and +0.78 there, and +1.99 in seed 2. The sign rule labels H02 and H05 on YOLO11s as "selection-independent". However, the `last.pt` Δs of these two cards are zero (+0.00, +0.03). Their magnitude at `best.pt` (+1.14, +2.28) depends on the selection. For these two cards, the label is a borderline case of the sign rule.

Conclusion.

The H01 acceptance is reported under the pre-specified rule. It carries the "sensitive to `best.pt` selection" label and a note that most of the gain comes from a single test scene group. No claim of a robust improvement is made. Measured: for YOLO11s, the `best.pt` and `last.pt` Δs diverge systematically. For YOLO26n, they diverge in the opposite direction. Remaining hypothesis: `best.pt` selection captures a peak specific to a small validation set of 10 scene groups. This peak partly does not carry over to the test set. The causes would be selection bias ("winner's curse") and scene-group noise. The experiment that would separate these explanations was not run. It would require fixed-epoch evaluation at intermediate epochs, or more seeds.

C.5 KITTI robustness analysis (post-unlock, exploratory)

These analyses were performed after the KITTI test set was opened. They involve no new test opening, inference, or training. The existing `best.pt` and `last.pt` test predictions were re-read, and the Δs recorded on the full sample were reproduced. These analyses are not part of the pre-specified rules, and no Holm correction was applied.

Table C5 reports a hierarchical bootstrap. Each replicate resamples with replacement, first the three seeds and then the 13 test scene groups. The group draw is shared by all cards and seeds (paired). The seed draw is over only $n = 3$, so the intervals may still be optimistic. The interval covers zero for 13 of the 14 cards. The YOLO11s H01 numbers are given in Section 7.2 of the main article. Only one card has an eight-class Δ interval that excludes zero (blue): YOLO11s H05, with [+0.20, +4.10]. However, H02 and H05 are directional replication cards, with an expected $\Delta < 0$. In addition, the `last.pt` Δ of H05 is ≈0. The blue cell is therefore not an improvement claim. The without-tram and 6-class columns show no interval, so they are not colored. For the without-tram Δ, the interval is [+0.11, +3.32] for YOLO11s H05 and [+0.07,

+3.34] for H08. The positive-group column is not image-weighted. Single-group Δs are unstable for small groups of 5–6 images.

The YOLO11s H01 acceptance has further sensitivities. The one-sided p of the clustered bootstrap is 0.0535. This is a sensitivity analysis, not a pre-specified test. The acceptance relies on the normality assumption of the t test. With three seeds, a sign-flip test cannot give a one-sided p below 0.125. The rule document does not provide for pooling across models. As a sensitivity analysis, we corrected the 10 improvement tests of the two models jointly with Holm-10, which gives $p = 0.0848$. Pooling all 14 tests into a single family gives $p = 0.119$. These values do not change the pre-specified decision. Leaving out drive_0056 lowers the Δ of H01 from +1.49 to +0.45, a drop of $\approx$70%. mAP is not additive over groups, so this is a leave-out difference. The gain does not appear in the standard three classes (car, pedestrian, cyclist). It comes from the classes with few samples (Table C1).

The clustered bootstrap CI measures only the sampling uncertainty of the test scene groups at fixed weights. It does not include seed variance. For the improvement cards that were not accepted, its upper bound is at most +2.92 for YOLO26n and +2.76 for YOLO11s. For YOLO11s, it excludes zero for five of seven cards (H03, H04, H05, H06, H08). For the accepted H01, in contrast, it crosses zero. For YOLO26n, it excludes zero for no card. This pattern points to a common difference that shifts all cards against the same baseline, not to a card-specific effect (Section C.4). It is not taken as evidence of a card effect.

Validation (987 images, 10 scene groups) is a secondary endpoint. It did not select the same card as the test set. For YOLO26n, there is no acceptance on validation either. For YOLO11s, only H04 met the acceptance criterion on validation, with +1.82 $\pm$ 0.43 and $p_{\text{Holm}} = 0.046$. This is not a decision. H01 was not accepted on validation, with +1.73 $\pm$ 1.78 and $p_{\text{Holm}} = 0.350$. On the test set, the picture reverses. H01 meets the criterion, whereas H04 does not ($+1.07 \pm 0.72$, $p_{\text{Holm}} = 0.252$). The two sets select different cards. This is consistent with the identity of the single accepted YOLO11s card being sensitive to scene-group sampling.

C.6 Details of training-free inference (H07)

H07 was run on validation with the baseline `best.pt` weights. For each model, the single row with the highest validation mean entered the test. For YOLO11s seed 2, these `best.pt` weights are the epoch-1 weights (Section C.4).

For YOLO26n, the selected U832 row gave +1.50 on validation and +1.54 $\pm$ 0.58 points on the test set. Latency at batch 1 rose from 7.74 ms to 7.80 ms, a ratio of 1.008. This is within the pre-specified $\times$1.10 budget. The pixel count grew 1.69-fold, yet latency increased by only 0.8%. This suggests that per-image Python and preprocessing overhead dominate the measurement. On VOC, the latency ratio of the same row was 1.14.

For YOLO11s, the selected W2 row is the WBF fusion of the T2 branches. It gave +1.61 on validation and +2.02 $\pm$ 0.95 on the test set. Latency rose from 6.82 ms to 26.70 ms ($\times$3.91), which is outside the pre-specified $\times$1.10 budget.

The two models use different latency engines. YOLO11s uses the VOC engine, and YOLO26n uses per-image `predict`. At R0, nano gives 7.74 ms and small gives 6.82 ms.

Absolute ms values are therefore not compared across models; only the within-model ratio is read. These Δs are measured with the same weights. They therefore do not include the `best.pt` selection difference between the cards and the baseline. However, they are single-row, three-seed, and exploratory, and no CI or Holm correction is given. On VOC validation, 832 px input had changed mAP50-95 by −5.19 (Section 5.2 of the main article). On KITTI, the same row has a positive sign. The untested hypothesis for this contrast is the resolution difference described above (Section C.1).

Latency measurement error and its correction.

The first run of the KITTI H07 script crashed with a CUDA out-of-memory error. This happened during the latency measurement of the selected row, and no latency value was written. The image paths had been passed to `predict` as a single source list. Ultralytics processes such a list in a single batch, so the batch size of 1 required by the rule document was not met. The error was corrected before any latency value was produced and before the KITTI test set was opened. For YOLO11s, the VOC H07 measurement engine was used unchanged. For YOLO26n, per-image `predict` with GPU synchronization was used. The 50 warm-up iterations, 300 measurements, and FP16 were unchanged. The definition of the accuracy rows and the selection rule were also unchanged. The reported KITTI latencies come from the corrected measurement. The VOC latencies were not affected.

Table C5 KITTI test robustness (post-unlock, exploratory): card − baseline Δ [points]. Hierarchical bootstrap over seeds and the 13 scene groups ($B = 2000$). Blue+bold: the CI excludes zero.

Card	Δ	Hierarchical CI95	$P(\Delta \leq 0)$	Δ w/o tram	6-class Δ	Pos. groups
YOLO26n						
H01 multi-scale	+0.12	[−2.26, +2.51]	0.401	+0.36	+0.11	5/13
H02 Inner-CIoU	−0.93	[−2.82, +0.85]	0.778	−0.28	−1.21	5/13
H03 box copy-paste	+1.01	[−0.81, +3.36]	0.093	+1.78	+1.20	7/13
H04 mosaic scale	−1.24	[−3.05, +1.65]	0.758	−0.38	−1.81	9/13
H05 Varifocal	+0.61	[−1.53, +2.32]	0.278	+0.25	+0.77	7/13
H06 box loss gain	−0.25	[−2.06, +1.54]	0.600	+0.09	−0.31	5/13
H08 high dose	+0.71	[−0.93, +2.73]	0.147	+1.24	+0.87	7/13
YOLO11s						
H01 multi-scale	+1.49	[−0.67, +2.61]	0.077	+0.53	+2.03	9/13
H02 Inner-CIoU	+1.14	[−0.76, +2.37]	0.120	+0.19	+1.53	10/13
H03 box copy-paste	+1.41	[−0.36, +3.50]	0.050	+1.20	+1.70	11/13
H04 mosaic scale	+1.07	[−1.00, +3.01]	0.116	+0.72	+0.96	11/13
H05 Varifocal	**+2.28**	**[+0.20, +4.10]**	**0.018**	+1.42	+3.15	9/13
H06 box loss gain	+0.57	[−0.81, +2.68]	0.150	+0.82	+0.68	10/13
H08 high dose	+1.68	[−0.45, +4.62]	0.084	+1.54	+1.78	11/13

$P(\Delta \leq 0)$: share of bootstrap draws with $\Delta \leq 0$. W/o tram: 7 classes. 6 classes: excluding person_sitting and misc. These two columns are uncolored (no CI is shown). Pos. groups: scene groups with seed-mean $\Delta > 0$. All values use `best.pt`. H02, H05: directional cards (expected $\Delta < 0$).

Appendix D Method details and computation notes

This section gives the details moved out of the Methods section of the main article. These are the hypothesis card and decision rules, the expansion of the loss terms, the training recipe and the intervention details. It also gives computation notes. The screening probabilities, the gradient-norm ratio and the sources of the plausibility anchor are in Appendix A of the main article.

D.1 Hypothesis card and decision rule details

In addition to the fields described in the main article, the card also fixes the post-acceptance ablations. The card file must be timestamped before the first run. In practice, the evidential strength of this rule is limited (Section 9 of the main article). A combined card can be written only if each of its components has been accepted separately. For cards for which no isolated VOC ablation could be found, the expected ranges are narrow and centered on zero.

A false elimination at screening is costly. The threshold was therefore kept below the single-seed variability. Even at zero effect, the pass probability is not negligible, so a pass requires a second confirmation. Some cards eliminated at screening were later completed to three seeds. For such a card, failing the KEEP criterion is described as a "hit", and meeting it as a "missed gain". A hit is a specificity measure. It is likely even under an uninformative gate, so it is not considered informative (Appendix A.4 of the main article). In the KEEP rule, s_Δ is the sample s.d. of the $\Delta_{k,s}$. The alternative of the one-sided test is $H_1 : \bar{\Delta}_k > 0$. In the INSPECT case, an additional seed is added only from the K5 budget, for both the variant and the baseline. The KEEP rule was frozen in round 2, after the H01–H03 results had been seen. A card that passes the round-2 rule ($\bar{\Delta}_k \geq +0.40$) also passes the round-1 rule, but not vice versa.

D.2 Expansion of the baseline loss terms

The neck of the baseline is a path aggregation network (PAN). It produces outputs at three scales (strides 8, 16 and 32); see the third-party overview [6]. In TAL, the class target of the positives is $t_i = a_{ij} \max_{i'} u_{i'j} / \max_{i'} a_{i'j}$. Eq. (3) of the main article uses CIoU [7] and the distribution focal loss (DFL) [8]. They are defined as follows:

$$\mathrm{CIoU} = \mathrm{IoU} - \frac{\rho^2(\hat{\mathbf{m}}, \mathbf{m})}{d^2} - \alpha_v v, \tag{D2}$$

$$\mathcal{L}_{\mathrm{dfl}} = \frac{1}{T} \sum_{i \in \mathcal{P}} w_i \, \frac{1}{4} \sum_{e=1}^{4} \Big[-\big((r_e + 1) - y_e\big) \log S_{e,r_e} - (y_e - r_e) \log S_{e,r_e+1} \Big]. \tag{D3}$$

In Eq. (D2), $\hat{\mathbf{m}}$ and $\mathbf{m}$ are the box centers and ρ is the Euclidean distance. d is the diagonal of the smallest box enclosing both boxes. The aspect-ratio term is $v = \frac{4}{\pi^2} \big(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{\hat{w}}{\hat{h}}\big)^2$, and $\alpha_v = v / \big((1 - \mathrm{IoU}) + v\big)$ [7]. BCE is $\mathrm{BCE}(z, t) = -t \log \sigma(z) - (1 - t) \log(1 - \sigma(z))$. The DFL in Eq. (D3) [8] predicts each of the four edge distances with a softmax distribution S_e over $R = 16$ discrete bins. Here y_e is the target distance and $r_e = \lfloor y_e \rfloor$.

D.3 Training recipe and data augmentation defaults

The transforms are applied in the following order. Mosaic, with probability 1.0, combines four images on a $2s \times 2s$ canvas ($s = 640$) [9]. RandomPerspective draws the scale factor $a \sim U[1 - \sigma_s, 1 + \sigma_s]$ with $\sigma_s = 0.5$ and applies a translation of 0.1. Rotation, shear and perspective are zero. The output is cropped to the central $s \times s$ window. Next come an HSV shift (hue-saturation-value; 0.015, 0.7, 0.4) and a horizontal flip with probability 0.5. MixUp and the built-in copy-paste are disabled. Mosaic is turned off in the last 10 epochs (`close_mosaic=10`). The built-in copy-paste requires masks, so it has no effect on the box-labeled VOC [10]. H03 and H08 therefore rely on a new box-based implementation.

D.4 Intervention details

Scale and box loss.

H01 aims to enforce scale invariance by showing the same object at different pixel densities. In H04, the RandomPerspective scale range is reduced from $[0.5, 1.5]$ to $[0.75, 1.25]$; the amplitude is not increased. In H02, the auxiliary box for each positive anchor is $b^{\text{in}} = (x_c, y_c, rw, rh)$. The DFL term, TAL and the gains are unchanged. The implementation was verified with an $r = 1$ identity unit test. In H05, the effective gradient norm is measured (Appendix A.5 of the main article).

Box-based copy-paste.

Copy-paste tries to close the small-object gap. It replicates small objects within the same image without overlap [11]. The transform runs after mosaic and before RandomPerspective. Candidates are objects that do not overlap other boxes (IoA ≤ 0.1). From these, $k \sim U\{1, 2, 3\}$ objects are selected and pasted with 3-pixel alpha blending. The steps are given in Section B.2. A maskless patch also carries a piece of background. The risk that the model learns the patch edge is one reason why AP_{75} is monitored. The transform is disabled in the last 10 epochs. H03 uses $p = 0.5$ and $\tau_A = 48$ pixels, and H08 uses $p = 1.0$ and $\tau_A = 96$ pixels. The paste thresholds are measured on the mosaic canvas. The $\text{AP}_S/\text{AP}_M/\text{AP}_L$ boundaries are measured in original pixels, following the COCO definition [12]. In H08, enlarging the pool also changes the size mix of the pasted objects. Probability and pool were therefore not separated in H08. The dose $D = N_{\text{pasted}}/N_{\text{mosaic}}$ [objects/mosaic] is measured approximately. It is derived from counters that the worker processes write periodically. The H08 card pre-specified a target dose of $D \geq 0.80$, with an expected range of 0.8–1.2. If the D measured before the run lies outside the gate [0.6, 1.5], the run does not start. A dose below the target is not adjusted, to avoid introducing a second variable. The realized dose is reported.

Training-free inference rows (H07).

The reference pipeline R0 uses a confidence threshold of 0.001 and an NMS IoU threshold of 0.7. It keeps at most 300 detections and uses a single scale of 640 pixels with class-wise classic NMS. Each row differs from R0 in only one component, and rows are not combined. *NMS threshold (N)* sets the IoU threshold to one of {0.5, 0.6, 0.8}. *Soft-NMS (S)* decays the score of an overlapping box per class with the Gaussian

rule, using $\sigma \in \{0.3, 0.5\}$ [3]. Its definition is given in Section B.3. *Test resolution (U)* increases the input size to 704, 768 or 832 pixels. *TTA (T)*: T1 is the Ultralytics default [13, 14]. It runs three branches with scales {1, 0.83, 0.67} and flips the middle branch horizontally. The branch outputs are passed to a single NMS in the original coordinates. In T2, the scale list is {1, 1.2, 1.4}. The three branches raise the total number of input pixels by a factor of $1 + 0.83^2 + 0.67^2 \approx 2.14$. Latency grows by this factor plus the cost of branch merging and NMS. *WBF (W)* passes the T1 or T2 branches through NMS separately and then merges them by fusion [15]. It uses an IoU threshold of 0.6, a skip threshold of 0.001 and average confidence. The latency ratio is computed relative to R0 measured in the same session. The threshold $\rho \leq 1.10$ was set before the run. It is the largest latency increase under which a real-time claim can still be made. GAIN rows do not enter the parent chain of the training cards.

D.5 Statistical details

This section expands on the "Statistics" paragraph in Section 3.3 of the main article. For training cards, seed pairing is only nominal. The intervention also changes how the data augmentation consumes the random number generator (RNG), so pairing reduces the variance only partially. The Holm correction [16] uses a K5 family of twelve tests: H03 and TTA, each × six metrics. This count includes H03 mAP50-95 and is conservative. Holm-7 is applied to the three-seed validation differences of the seven cards. The image-level paired bootstrap [17] applies the same image indices to the variant and the baseline. It uses $B = 1000$ and a 95% percentile interval on the mean Δ over the three seeds. This interval does not include training-seed variance. For training interventions, we therefore also give a seed-level interval $\bar{\Delta}_k \pm t_{0.975,2}\, s_\Delta/\sqrt{n}$, with $t_{0.975,2} = 4.303$. Uncorrected $p < 0.05$ is not color-coded in the tables. The practical margins (+0.30 and +0.40 points) are a criterion separate from the statistical one. In K5, the test mAP50-95 of the baseline is compared, for each seed, with the value from the plausibility check. Any discrepancy is reported without correction.

D.6 External reference value

Only one external reference value is available for the order of magnitude of the KITTI baseline. It is a community model card (YOLO11n, KITTI; https://huggingface.co/dronefreak/kitti-yolo11n, accessed 2026-10-22). The card reports 24.0 for validation mAP50-95 and 67.3 for car AP50-95. The model was trained at 1280 px on the 3712/3769 split. The handling of `DontCare` is not specified. Early stopping (epoch 43) and the reported numbers rely on the same validation set. Resolution, split and evaluation differ, so these values cannot be compared with ours. The model card is not a peer-reviewed source.

References

[1] Zhang, H., Xu, C., Zhang, S.: Inner-IoU: More Effective Intersection over Union Loss with Auxiliary Bounding Box. arXiv:2311.02877 (2023)

[2] Dwibedi, D., Misra, I., Hebert, M.: Cut, paste and learn: Surprisingly easy synthesis for instance detection. In: 2017 IEEE International Conference on Computer Vision (ICCV), pp. 1310–1319. IEEE, Piscataway, NJ, USA (2017). https://doi.org/10.1109/ICCV.2017.146

[3] Bodla, N., Singh, B., Chellappa, R., Davis, L.S.: Soft-NMS – improving object detection with one line of code. In: 2017 IEEE International Conference on Computer Vision (ICCV), pp. 5562–5570. IEEE, Piscataway, NJ, USA (2017). https://doi.org/10.1109/ICCV.2017.593

[4] Bolya, D., Foley, S., Hays, J., Hoffman, J.: TIDE: A general toolbox for identifying object detection errors. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, pp. 558–573. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58580-8_33

[5] Kumar, A., Raghunathan, A., Jones, R., Ma, T., Liang, P.: Fine-tuning can distort pretrained features and underperform out-of-distribution. In: International Conference on Learning Representations (ICLR) (2022). https://arxiv.org/abs/2202.10054

[6] Khanam, R., Hussain, M.: YOLOv11: An Overview of the Key Architectural Enhancements (2024). https://doi.org/10.48550/arXiv.2410.17725

[7] Zheng, Z., Wang, P., Liu, W., Li, J., Ye, R., Ren, D.: Distance-IoU loss: Faster and better learning for bounding box regression. Proceedings of the AAAI Conference on Artificial Intelligence **34**(07), 12993–13000 (2020) https://doi.org/10.1609/aaai.v34i07.6999

[8] Li, X., Wang, W., Wu, L., Chen, S., Hu, X., Li, J., Tang, J., Yang, J.: Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection. In: Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 21002–21012. Curran Associates, Inc., Red Hook, NY, USA (2020). https://proceedings.neurips.cc/paper_files/paper/2020/file/f0bda020d2470f2e74990a07a607ebd9-Paper.pdf

[9] Bochkovskiy, A., Wang, C.-Y., Liao, H.-Y.M.: YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv:2004.10934 (2020)

[10] Ultralytics: Data Augmentation using Ultralytics YOLO. https://docs.ultralytics.com/guides/yolo-data-augmentation/. Online documentation. Accessed: 2026-10-01 (2026)

[11] Kisantal, M., Wojna, Z., Murawski, J., Naruniec, J., Cho, K.: Augmentation for Small Object Detection. arXiv:1902.07296 (2019)

[12] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft COCO: Common objects in context. In: Computer

Vision – ECCV 2014. Lecture Notes in Computer Science, pp. 740–755. Springer, Cham (2014). https://doi.org/10.1007/978-3-319-10602-1_48

[13] Ultralytics: Test-Time Augmentation (TTA) — YOLOv5 documentation. https://docs.ultralytics.com/yolov5/tutorials/test_time_augmentation/. Online documentation. Accessed: 2026-09-29 (2026)

[14] Ultralytics: ultralytics v8.4.165 (release tag). https://github.com/ultralytics/ultralytics/releases/tag/v8.4.165. Software, version 8.4.165, commit 004db34b12f65fa5a160de2b1090f6943106bd35. Accessed: 2026-09-29 (2026)

[15] Solovyev, R., Wang, W., Gabruseva, T.: Weighted boxes fusion: Ensembling boxes from different object detection models. Image and Vision Computing **107**, 104117 (2021) https://doi.org/10.1016/j.imavis.2021.104117

[16] Holm, S.: A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics **6**(2), 65–70 (1979)

[17] Efron, B.: Bootstrap methods: Another look at the jackknife. The Annals of Statistics **7**(1), 1–26 (1979) https://doi.org/10.1214/aos/1176344552